Test Plan for Automation Exercise Web Application

Black-and-White Version | Based on the Official Automation Exercise Test Cases

Project Name: Automation Exercise Demo E-commerce Website

Test Plan ID: TP-AE-001

Prepared By: Group 2

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Tested URL: https://automationexercise.com

Version: 2.0 (Black Only / Official TC-aligned)

1. Introduction

This test plan defines the scope, objectives, environment, test basis, deliverables, and execution approach for validating the Automation Exercise web application. The document is aligned with the official Automation Exercise UI test cases page and the official API list page. The goal is to organize the testing work in a professional format while keeping the feature coverage consistent with the website's official practice scenarios.

2. Test Basis

The main test basis for this document is:

- Official Automation Exercise UI Test Cases page (TC-01 to TC-26).
- Official Automation Exercise API Testing list (API 1 to API 14).

3. Objectives

- Verify the core e-commerce user journeys defined by the official website test cases.
- Ensure the main UI flows work correctly across supported browsers.
- Validate navigation, forms, product features, cart behavior, checkout, and order confirmation.
- Validate selected public API endpoints exposed by Automation Exercise.
- Provide a structured QA document that can be submitted as a practice or training deliverable.

4. Scope

- In Scope:
    - All official UI practice scenarios from TC-01 to TC-26.
    - All official API practice scenarios from API 1 to API 14.
    - **Authentication Component:** User Registration, Login, Logout, and Account Deletion.
    - **Product Catalog Component:** Product listing, Categories (Women, Men, Kids), Brands filtering (Polo, H&M, Biba, etc.), and individual product detail pages.
    - **Cart Component:** Adding/removing items, updating quantities, and verifying price calculations.
    - **Checkout Component:** Address validation, order placement, and invoice generation.
    - **API Component:** Verification of all documented CRUD operations (GET, POST, PUT, DELETE) provided on the API Testing page.
    - **Static Pages & Forms:** Contact Us form and Newsletter Subscription.
    - Smoke, functional, negative, exploratory, and compatibility checks for critical user journeys.
- Out of Scope:
    - Database validation.
    - Security penetration testing.
    - Server-side log validation.
    - Payment Gateway Integration (mocked on this platform).
    - Load and Stress Testing.

5. Official UI Test Cases Coverage

The following list is aligned with the official test case titles published on the website:

ID	Official Test Case Title
TC-01	Register User
TC-02	Login User with correct email and password
TC-03	Login User with incorrect email and password
TC-04	Logout User
TC-05	Register User with existing email
TC-06	Contact Us Form
TC-07	Verify Test Cases Page
TC-08	Verify All Products and product detail page
TC-09	Search Product
TC-10	Verify Subscription in home page
TC-11	Verify Subscription in Cart page
TC-12	Add Products in Cart
TC-13	Verify Product quantity in Cart

TC-14	Place Order: Register while Checkout
TC-15	Place Order: Register before Checkout
TC-16	Place Order: Login before Checkout
TC-17	Remove Products From Cart
TC-18	View Category Products
TC-19	View & Cart Brand Products
TC-20	Search Products and Verify Cart After Login
TC-21	Add review on product
TC-22	Add to cart from Recommended items
TC-23	Verify address details in checkout page
TC-24	Download Invoice after purchase order
TC-25	Verify Scroll Up using 'Arrow' button and Scroll Down functionality
TC-26	Verify Scroll Up without 'Arrow' button and Scroll Down functionality

6. Official API Coverage

The following APIs are included as the official API practice list:

API ID	Official API Scenario
API-01	API 1: Get All Products List
API-02	API 2: POST To All Products List
API-03	API 3: Get All Brands List
API-04	API 4: PUT To All Brands List
API-05	API 5: POST To Search Product
API-06	API 6: POST To Search Product without search_product parameter
API-07	API 7: POST To Verify Login with valid details
API-08	API 8: POST To Verify Login without email parameter
API-09	API 9: DELETE To Verify Login
API-10	API 10: POST To Verify Login with invalid details
API-11	API 11: POST To Create/Register User Account
API-12	API 12: DELETE METHOD To Delete User Account
API-13	API 13: PUT METHOD To Update User Account
API-14	API 14: GET user account detail by email

- ## 7. Test Strategy & Approach
- A shift-left approach will be utilized, combining manual exploratory testing to identify edge cases with a robust automated regression suite to ensure continuous stability.

7.1 UI Testing

- Functional testing of the official UI scenarios.
- Negative testing for invalid inputs and error handling.
- Compatibility testing on Chrome, Firefox, and Edge.
- Exploratory testing around the official user flows.
- Retesting and regression for resolved defects.

7.2 API Testing

- Validate response codes, response messages, and returned data structures.
- Check missing parameter behavior where defined by the official API list.
- Verify login and user account endpoints with valid and invalid data.

- **7.3 Automation Architecture**
  - **Framework:** Playwright or Selenium combined with a Page Object Model (POM) design pattern. Playwright is highly recommended for this specific site to handle the dynamic cart modals and parallel execution across Chromium, Firefox, and WebKit engines seamlessly.
  - **API Testing:** Integration of API requests within the automated suite to quickly seed test data (e.g., creating a user via API before running a UI checkout test) to reduce execution time.
  - **Defect Management:** Jira will be used for bug tracking. All identified defects will follow a strict workflow: *To Do -> In Progress -> Ready for QA -> Done*.

- **7.4 Testing Types**
  - **Functional Testing:** Validating each feature against expected behavior.
  - **Integration Testing:** Ensuring seamless data flow between the Cart, Checkout, and User Profile components.
  - **End-to-End (E2E) Testing:** Simulating complete user journeys from landing on the homepage to successfully placing an order.
  - **Cross-Browser Testing:** Validating UI consistency across major browsers.

Detailed E2E Test Scenarios

Component 1: User Authentication

- **TC-AUTH-01:** Register a new user with valid data and verify successful account creation.
- **TC-AUTH-02:** Attempt registration with an already existing email address and verify the error message.
- **TC-AUTH-03:** Login with valid credentials and verify the UI updates to show the logged-in state.
- **TC-AUTH-04:** Attempt login with incorrect credentials (invalid email/password combination).
- **TC-AUTH-05:** Verify the "Delete Account" functionality completely removes the user data.

- **Component 2: Product Catalog & Search**
  - **TC-CAT-01:** Verify visibility of all products on the main Products page.
  - **TC-CAT-02:** Navigate through Categories (e.g., Women > Dress) and verify the catalog updates to show only relevant items.
  - **TC-CAT-03:** Filter products by Brands (e.g., selecting "Madame" or "Mast & Harbour") and assert the results.
  - **TC-CAT-04:** Search for a specific product using the search bar and verify the accuracy of the results.
  - **TC-CAT-05:** Open a product details page and verify the presence of product name, category, price, availability, condition, and brand.

- **Component 3: Cart Management**
  - **TC-CRT-01:** Add a single product to the cart and verify the "Added! Your product has been added to cart" modal appears.
  - **TC-CRT-02:** Add multiple different products and verify all items are present in the "View Cart" page.
  - **TC-CRT-03:** Increase the quantity of a specific item in the cart and verify the total price updates correctly.
  - **TC-CRT-04:** Remove an item from the cart and verify it disappears and the total price recalculates.

- **Component 4: Checkout Process (The Critical E2E Path)**
  - **TC-CHK-01:** Complete the checkout process as an existing, logged-in user.
  - **TC-CHK-02:** Complete the checkout process by adding items as a guest, then registering during the checkout flow.
  - **TC-CHK-03:** Verify the delivery address and billing address display accurately on the checkout review page.
  - **TC-CHK-04:** Enter mock payment details, confirm the order, and verify the "Order Placed" success screen.
  - **TC-CHK-05:** Download the invoice post-purchase and verify the file format and content.

- **Component 5: API endpoints**
  - **TC-API-01:** Send a GET request to /api/productsList and verify a 200 OK status with the correct JSON payload.
  - **TC-API-02:** Send a POST request to create a user and verify the backend response.
  - **TC-API-03:** Send a DELETE request to /api/verifyLogin and ensure appropriate security/error codes are returned for unsupported methods.

8. Test Environment

Item	Description
Website URL	https://automationexercise.com
Browsers	Chrome (latest), Firefox (latest), Edge (latest)
OS	Windows 11
Tools	Postman, Selenium, browser DevTools, Excel / Word for reporting
Environment Type	Public demo / practice environment

9. Entry and Exit Criteria

9.1 Entry Criteria

- The website is reachable and usable.
- The API endpoints are accessible.
- Test data and required accounts are prepared.
- Browsers and tools are ready for execution.
- The testing environment is stable and accessible. All test scenarios are documented and approved. Automation framework is configured.

9.2 Exit Criteria

- All planned UI and API scenarios are executed.
- Critical and major issues are logged and reviewed.
- Retesting is completed for resolved issues.
- Test summary and evidence are prepared.

10. Deliverables

- **Test Plan Document:** This document, defining the strategy and scope.
- **Test Scenarios & Checklists:** High-level outlines of test conditions.
- **Automated Test Scripts:** The complete repository of automated checks covering UI and API components.
- **Defect Reports:** Documented and tracked issues within Jira.
- **Test Execution Summary:** A final report detailing test coverage, pass/fail metrics, and overall platform quality metrics.
- API collection (optional)
- Execution report and final summary
- 100% of planned test cases are executed. All critical and high-severity bugs are resolved and retested in Jira. Test summary report is generated.

11. Risks and Mitigation

Risk	Mitigation
Demo site instability	Retry blocked cases and document environment-related failures.
Dynamic data reset	Recreate accounts and test data when needed.
Unexpected UI popups or ads	Use stable waits and retry steps if required.
Duplicate email during signup	Use unique email addresses for registration tests.
Test Environment Instability or API Rate Limiting	Implement mock APIs for isolated UI testing of individual components. Maintain a fallback manual testing checklist for the most critical End-to-End (E2E) paths to ensure coverage if automation fails.
Test Data Corruption & State Dependency	Ensure all automated scripts are stateless. Implement teardown hooks in the automation framework to clean up data (e.g., deleting a registered user) post-execution, or utilize dynamically generated, unique test data for every test run.
Dynamic Element Locators Changing	Utilize resilient selector strategies in Playwright/Selenium (e.g., data-testid attributes where possible, or robust relative locators) rather than fragile XPath structures to minimize maintenance overhead.
API response changes	**Recheck expected responses before full execution.**

12.Defect Management & Triage Strategy

All identified defects will be logged in Jira and triaged based on the following severity matrix to prioritize developer remediation:

- **Blocker (Severity 1):** Critical failures preventing core functionality. (e.g., Users cannot add items to the cart; Checkout API returns 500 Internal Server Error; the application crashes). Testing on the affected component halts immediately.
- **Major (Severity 2):** Core functionality is impaired but a workaround exists, or non-critical components fail. (e.g., Product filtering returns incorrect brands; user profile fails to update).
- **Minor (Severity 3):** UI/UX issues, aesthetic inconsistencies, or minor functional glitches that do not impact the user journey. (e.g., Misaligned buttons, typos on the "Contact Us" page, color contrast issues).

13. Advanced Automation & Execution Strategy

To maximize test coverage and efficiency, the following advanced methodologies will be employed during the execution phase:

- **Data-Driven Testing (DDT): Test scripts for authentication, search, and checkout components will be designed using a data-driven approach. Scripts will ingest external data sets (JSON/CSV) containing arrays of valid, invalid, and boundary-value inputs to execute multiple scenarios through a single, optimized script.**
- **CI/CD Pipeline Integration: The automated regression suite will be integrated into the CI/CD pipeline (e.g., GitHub Actions/Jenkins). Tests are configured to trigger automatically upon new Pull Requests, ensuring zero regression in existing components before code is merged into the main branch.**
- **Edge-Case & Logic Break Validation: Manual exploratory sessions and targeted automation will stress the application's underlying logic. Scenarios include:**
    - **State Tampering: Authenticating a user in one browser tab, logging out in a second tab, and attempting to submit a checkout payload in the first tab.**
    - **Payload Injection: Intercepting API checkout requests to inject negative pricing values or invalid data types to validate backend sanitization.**
    - **Concurrency Checks: Simulating simultaneous cart additions for products with limited inventory.**

14. References

- Automation Exercise main website
- Automation Exercise official Test Cases page
- Automation Exercise official API list page

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